Data & AI Builders Conference

Building modern data foundations for intelligent solutions together

Mastering Fabric Data Agents

From setup to success



THANK YOU!!

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Lodestar
IT'S TIME TO INNOVATE

Collaborators



aspitalia.com

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Personas

Customer | Financial Advisors | Portfolio Managers | ...

Chat with your Data | Predictions

Data Management Platform



Fabric

Platforms

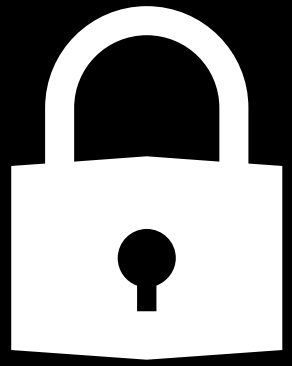
Digital Experience | CRM & ERP | Document Management | ...

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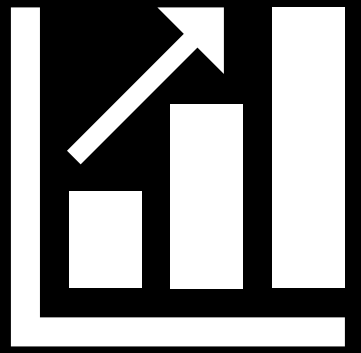
Let’s Define Success



Reliable answers



Secured



Well adopted

Marc Lelijveld

Architect & Technical Advocate
Macaw Netherlands

Microsoft® Most Valuable Professional

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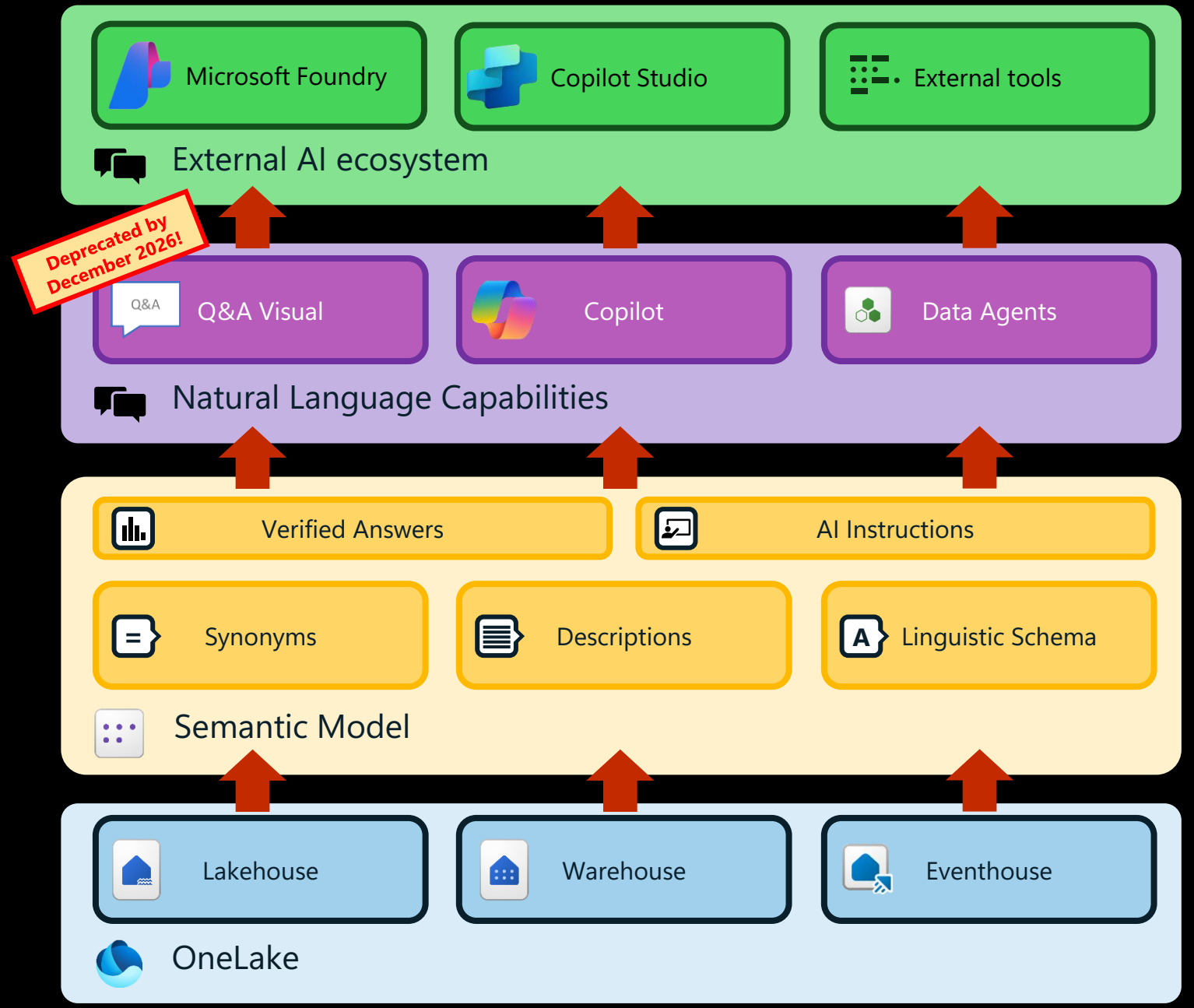
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DutchFabricUsergroup.com

FAVORITE STUFF:



Where does natural language interactions fit in?



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Exploring solution areas
What are the alternatives to copilot?

What can you conclude from the orders placed versus the orders shipped?

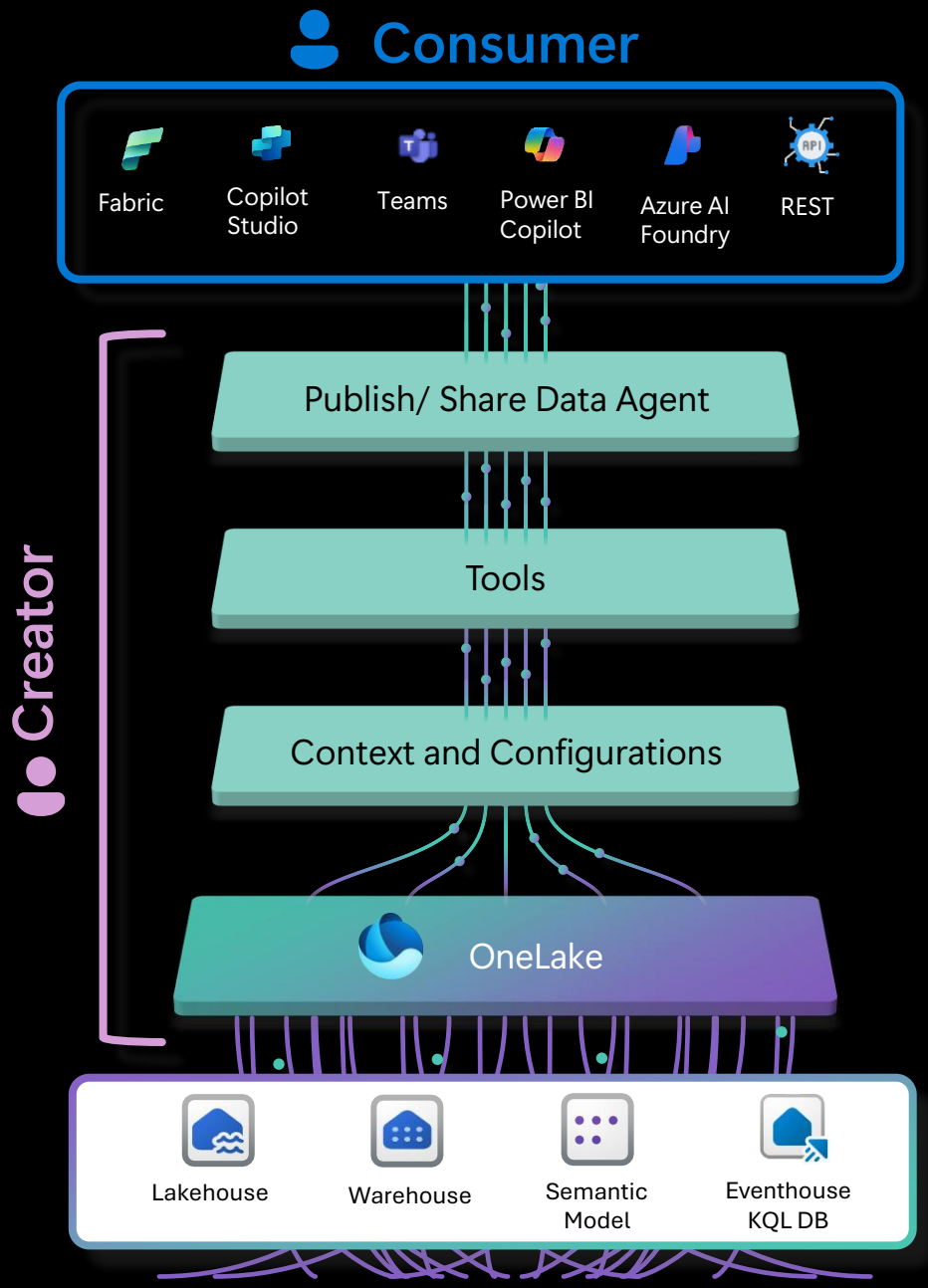




Understanding
Data Agents

Introducing Data Agents

- The **conversational Data Agent** allows users to interact with data naturally, enhancing accessibility and usability.
- Improved chat canvas for creators with **new debugging capabilities**, making it easier to understand and refine responses.
- Seamlessly reason over multiple data sources, including **Lakehouses, warehouses, semantic models and Eventhouse KQL databases**.
- Your Data Agent can be consumed inside and outside of Fabric. Stay tuned for **integrations** with Copilot Studio, Teams, Azure AI Foundry and your own custom applications.



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Prerequisites for Data Agents

- Fabric paid capacity (any size)
- Fabric Data Agent tenant setting is enabled
- Copilot tenant switch is enabled
- A valid source (Warehouse, Lakehouse, Power BI semantic model or KQL Database)
- Power BI semantic models via XMLA endpoints tenant switch is enabled.

Different types of supported sources



Lakehouse



Warehouse



Eventhouse

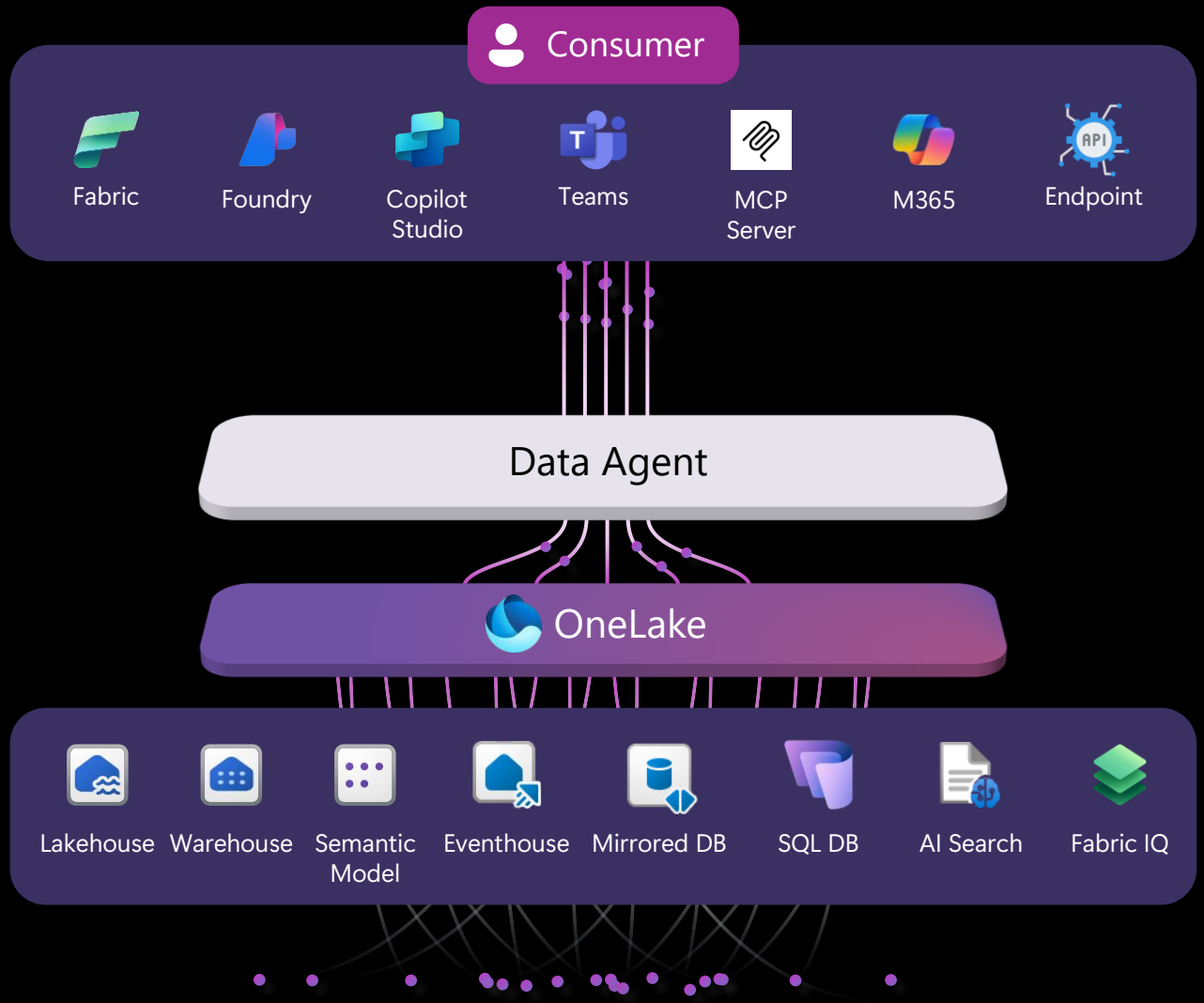


Semantic Model

And related sources (like mirrored sources)

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Fabric Data Agents



A **virtual analyst** that allows users to interact with and gain insights from enterprise data in OneLake

New Consume your data agents in **M365 Copilot** and as **remote MCP Server in VS Code**

New Data agents now support unstructured data through **Azure AI Search**

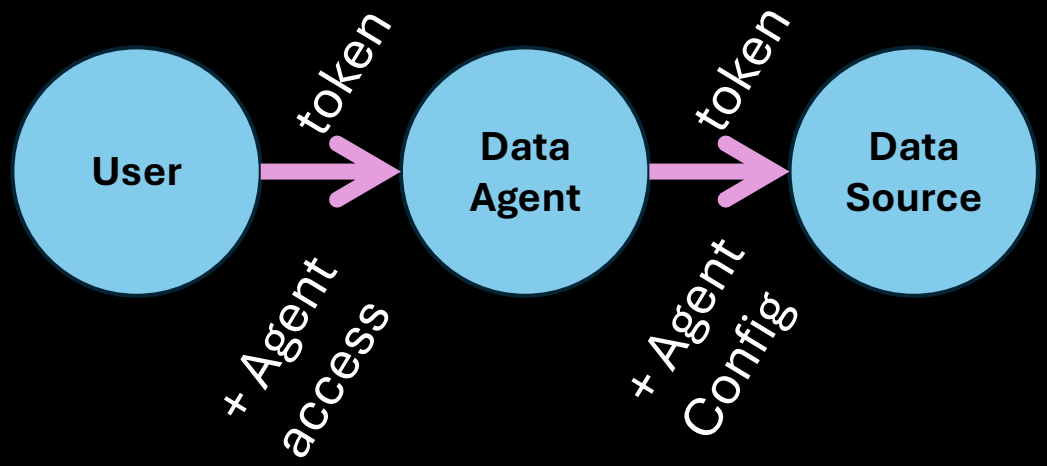
New Data agents now support **Fabric IQ Workload through Ontology**

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Security & Permissions – Data Sources

Pass-through mechanics

- User credentials flow to the data source(s)
- Permission evaluated per-user, per-query
- RLS/OLS etc. are baked-in and are inherited from the source(s)



Security & Permissions – Data Sources

Source Type	Minimum Permission	Reasoning
Semantic Model	Read (new!, previously build)	Agent generates DAX queries
Lakehouse	Read on the lakehouse item (and table access if enforced)	Write not required unless modifying data.
Warehouse	Read (SELECT on relevant tables)	Higher permissions only for DML/DDL operations.
KQL database	Reader role on the database	Higher roles only for management commands.
Ontology	Read on the ontology item, and Read on the underlying semantic model, lakehouse, or KQL database bound to the ontology	
Other supported sources	Query/read-level access	Must allow metadata + data retrieval.

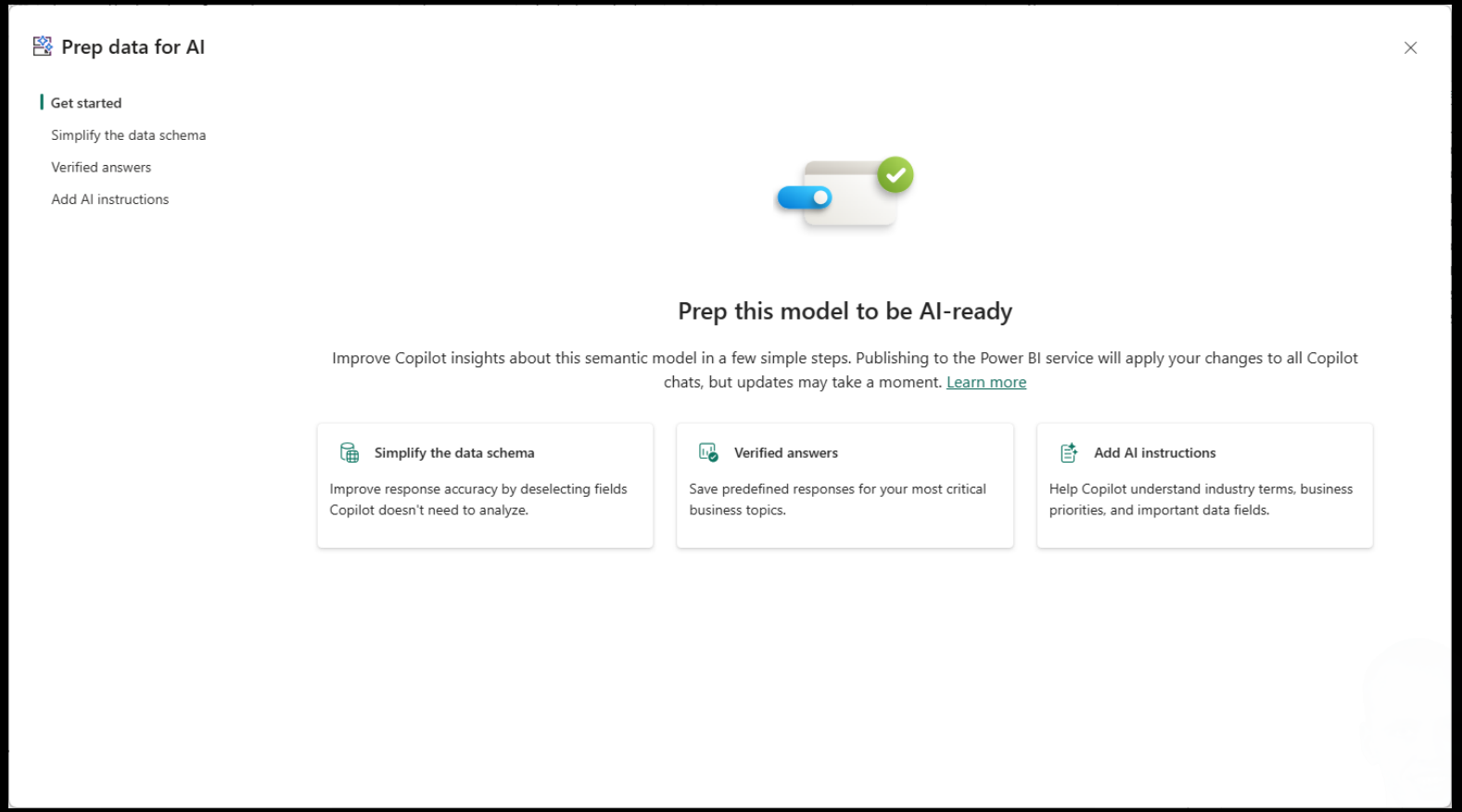


Semantic Models &
Prep Data for AI

Semantic models: Prep data for AI

Three main pillars

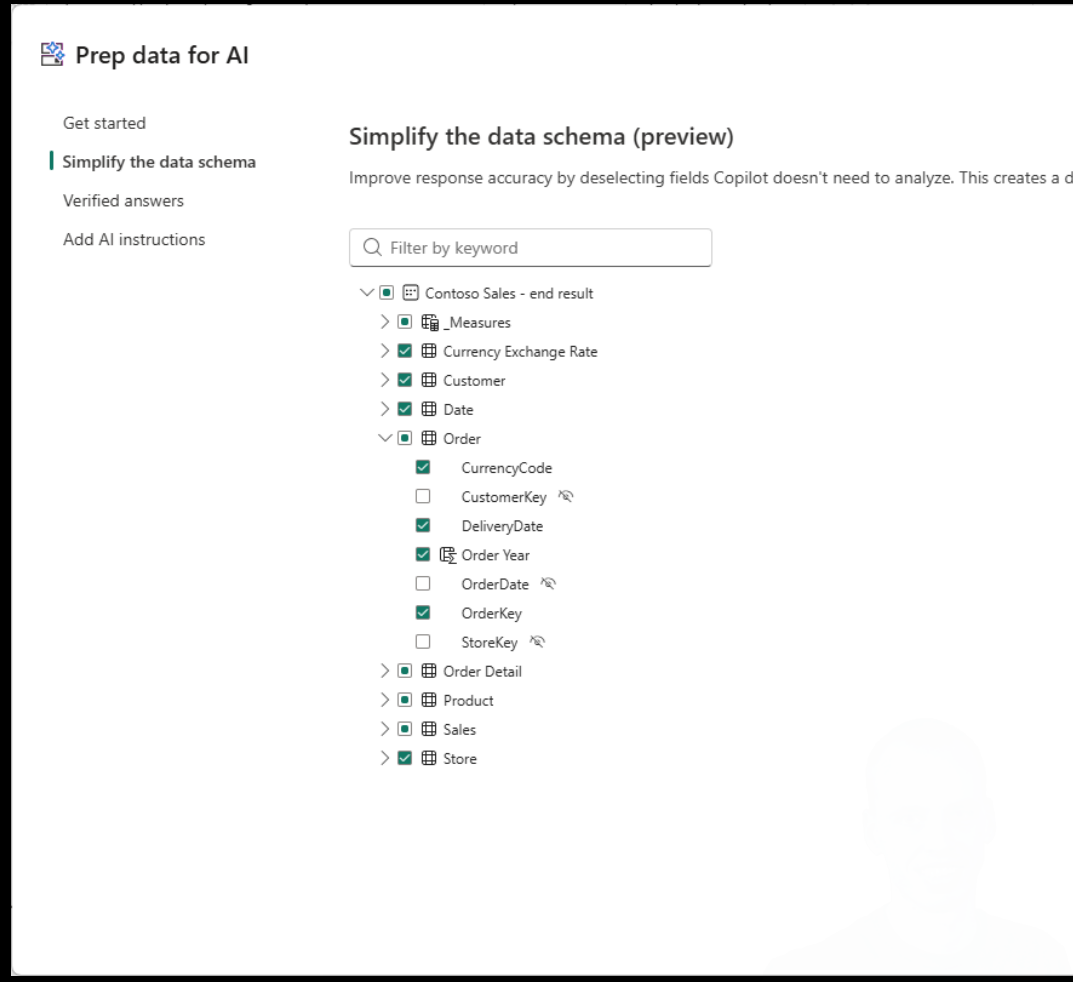
- Simplify schema
- Verified answers (links to visuals)
- AI Instructions



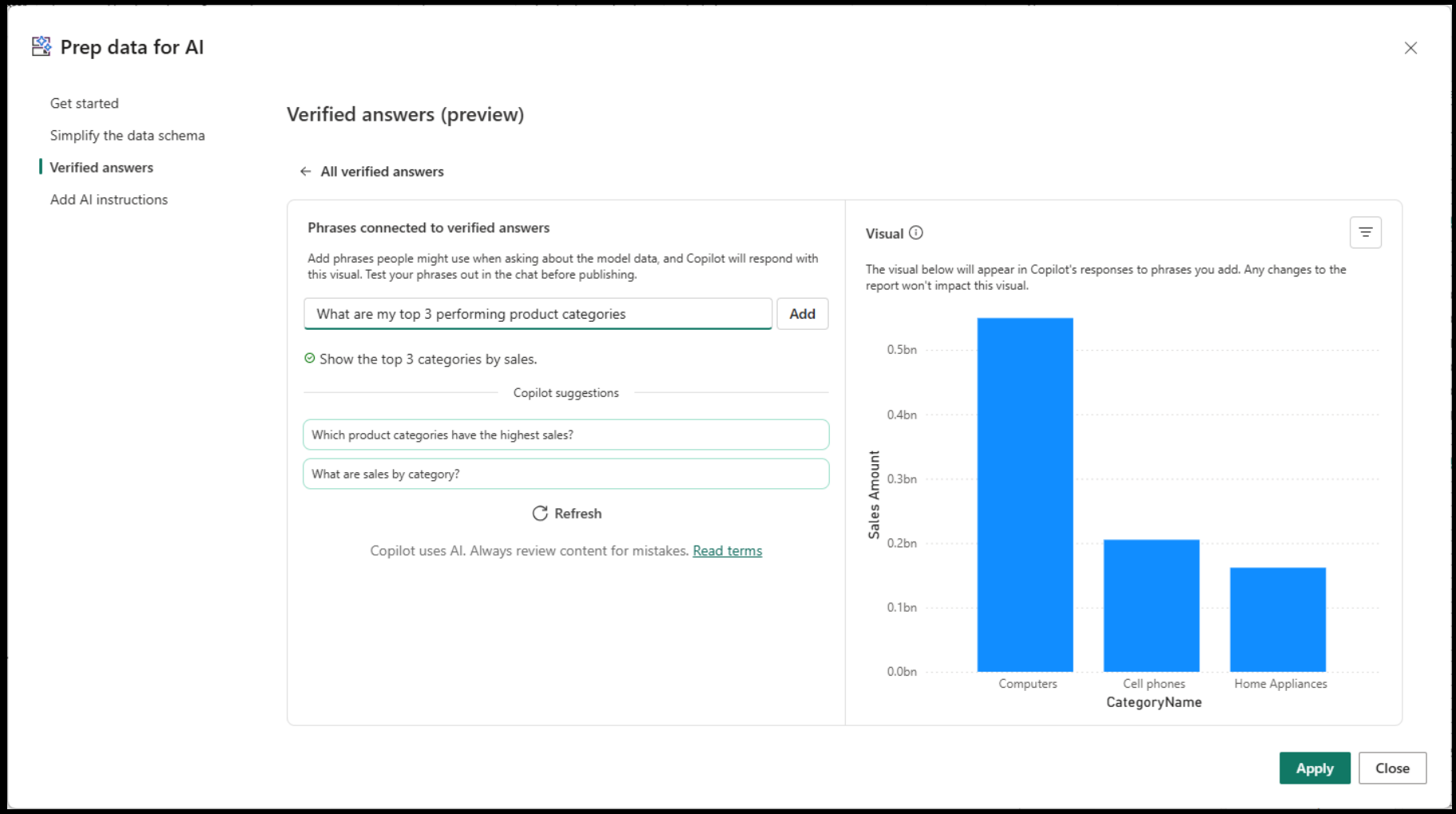
Semantic models: Prep data for AI – Simplify schema

- Only keep what is relevant!
- Hide report-specific measures
  - Conditional formatting
  - Measures outputting texts like visual titles
- Disconnected tables likely cause issues
  → build relationships!

Keep in mind that Visual Calculations are not part of the semantic model!



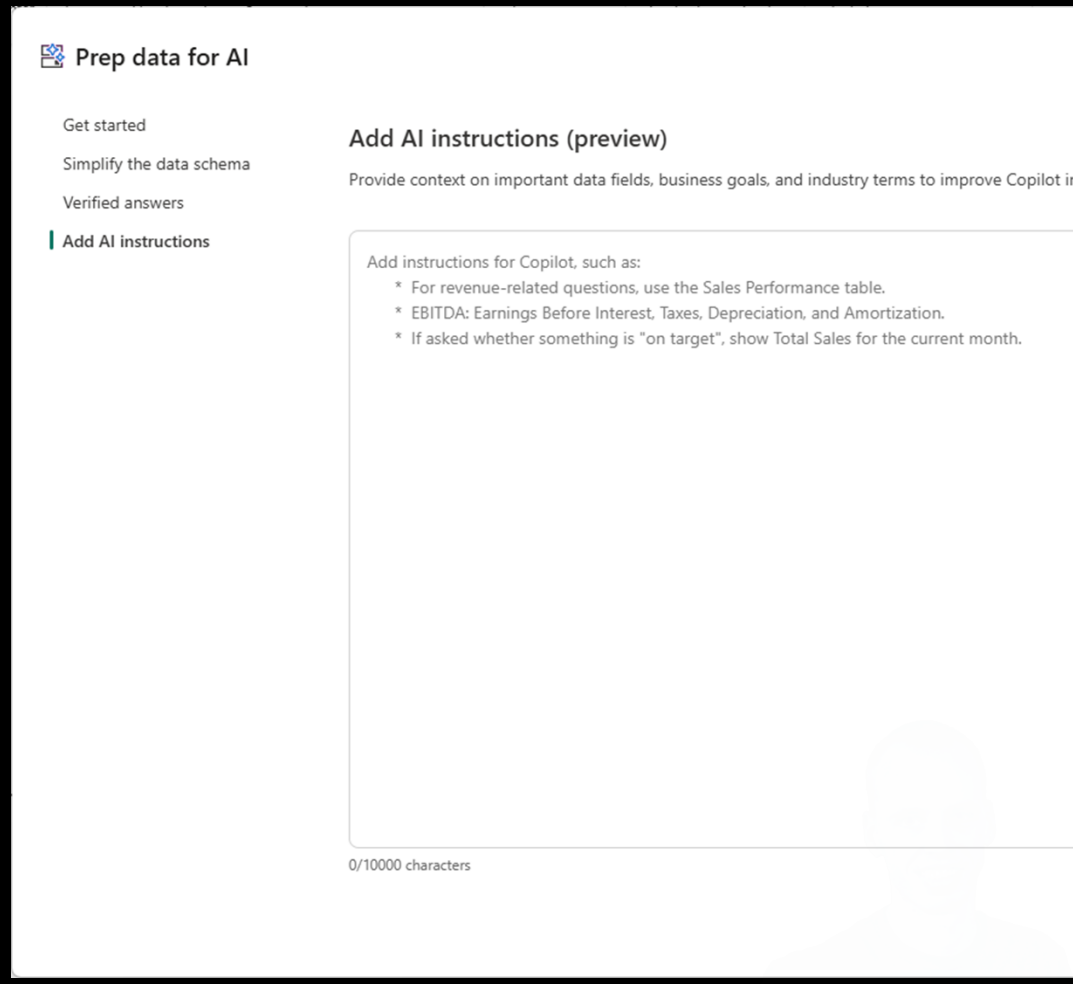
Semantic models: Prep data for AI – Verified answers



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Semantic models: Prep data for AI – AI instructions

- Unstructured guidance (textual)
- No guarantee the guidance is followed exactly – see it as a hint!
- Response formatting
  - Style
  - Tone
- Business specific logic – for example:
  - Define what is a top performer – high or low?
  - Peak / off-season: which months are these?



Common pitfalls

- Conflicting or overly **complex** instructions
- Skipping **schema refinement** – especially with large semantic models containing many tables
- Not using a **star schema** (resulting in inefficient DAX)
- Relying on **hidden fields**
- Including **unnecessary measures**: Simplify the schema,
  - Remove measures that are not relevant or broken,
  - Hide report measure (for conditional formatting for example),
  - Avoid duplicating logic in different measures

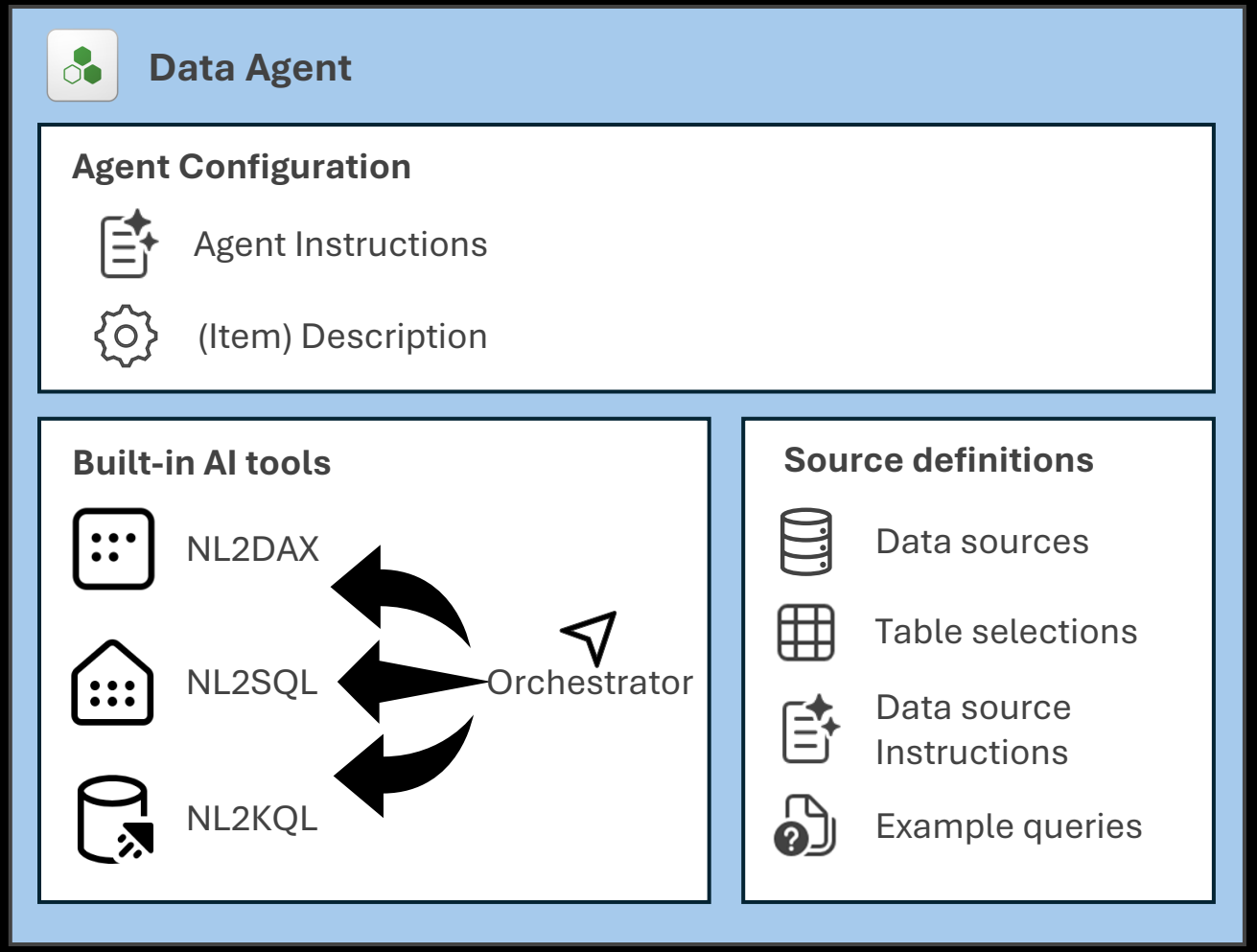
  Non-descriptive naming (technical names for objects)
- Relying on **implicit measures**: high chance of inconsistency!

  Ambiguous date fields (e.g. Order Date, Ship Date, …)



Data Agents
Innerworkings

Data Agent building blocks



Creator configurations



Instructions (schema)

Verified answers

Agent Instructions

Data Source Instructions

Example Queries

Intended Use Cases:

- Specify which data sources to route for specific questions
- Guide how the agent interprets user intent

What to include:

- Map terminology to user intent
- Define steps for handling key questions
- Set response style and tone

Intended Use Cases:

- Provide clarity on how to use each source
- Guide table selection for topics
- Ensure correct query logic

What to include:

- Descriptions of key tables & columns
- Recommended table usage for certain topics
- Example column values and filter patterns

Intended Use Cases:

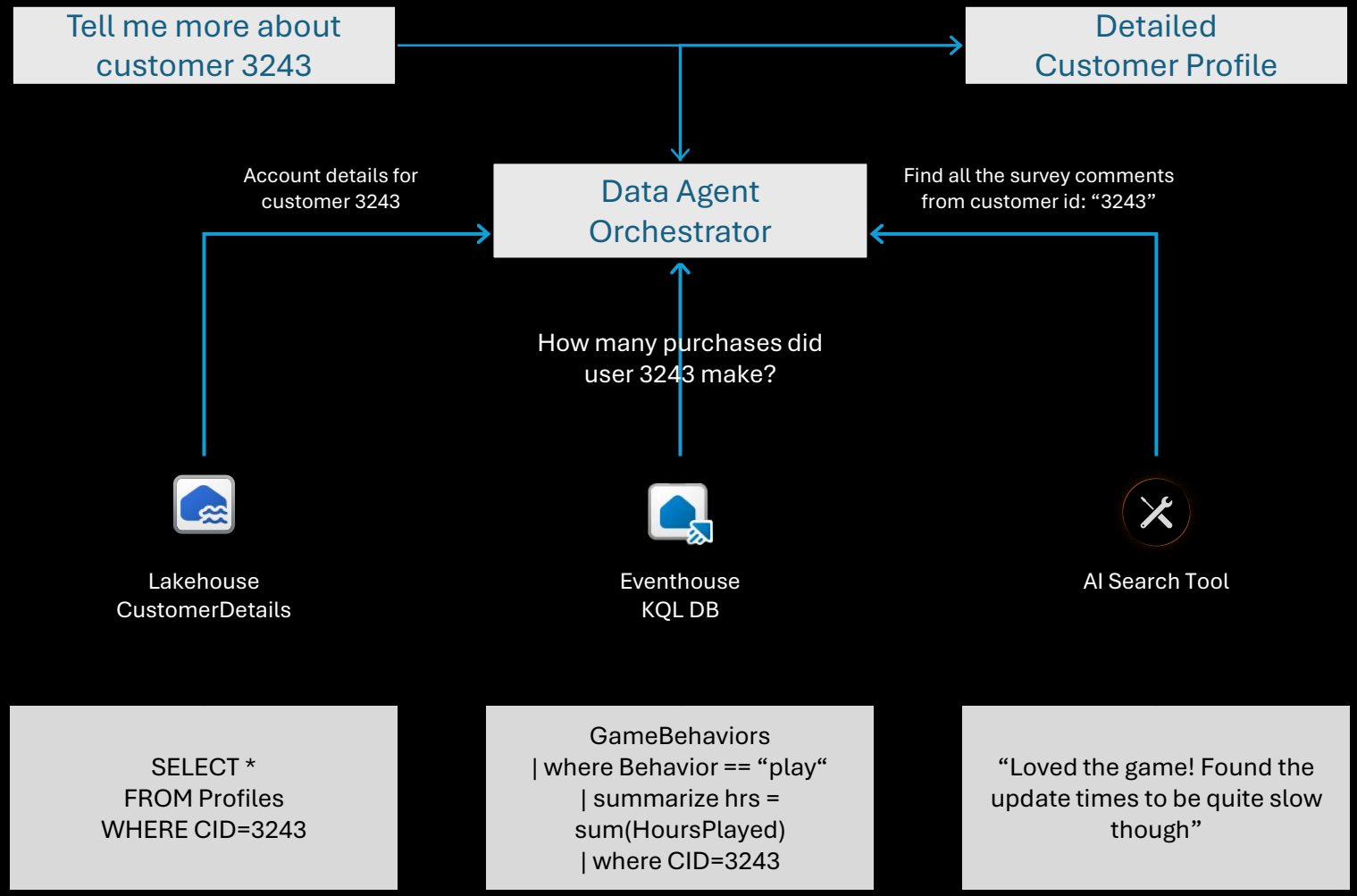
- Reflect advanced logic
- Examples should highlight join patterns or expected values

What to include:

- Representative set of questions a consumer may ask

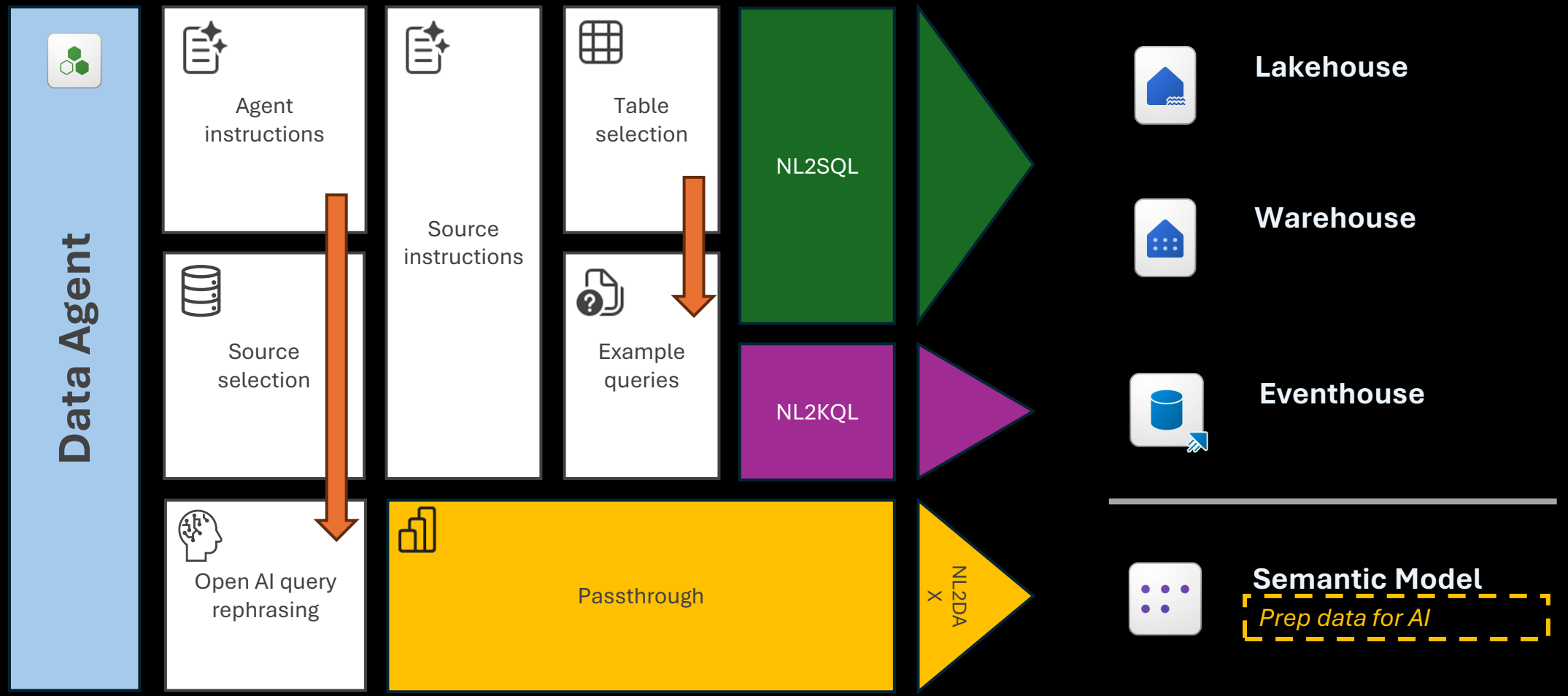
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The Data Agent flow

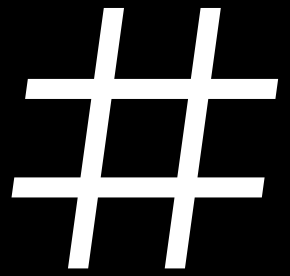


1. The data agent delegates questions to specific tools or data sources.
2. Data Sources are specialized at taking a natural language question and generating a response – query or result.
3. Data Sources & Tools supply their response back to the orchestrator – allowing it to process the information and think about the next step or return the summarized result.

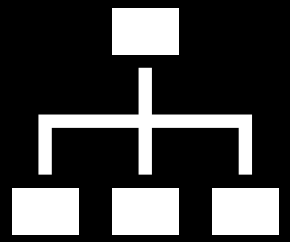
Data Agent Orchestrator - Inner workings



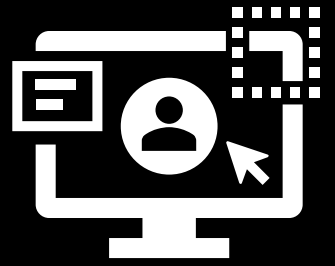
Advantages of using a Semantic Model as a data source



Allows metadata enrichment



Business logic (measures)



User friendly and terminology



Single source of truth (incl reports)

Keep in mind!

The answers will only be as good as your data!

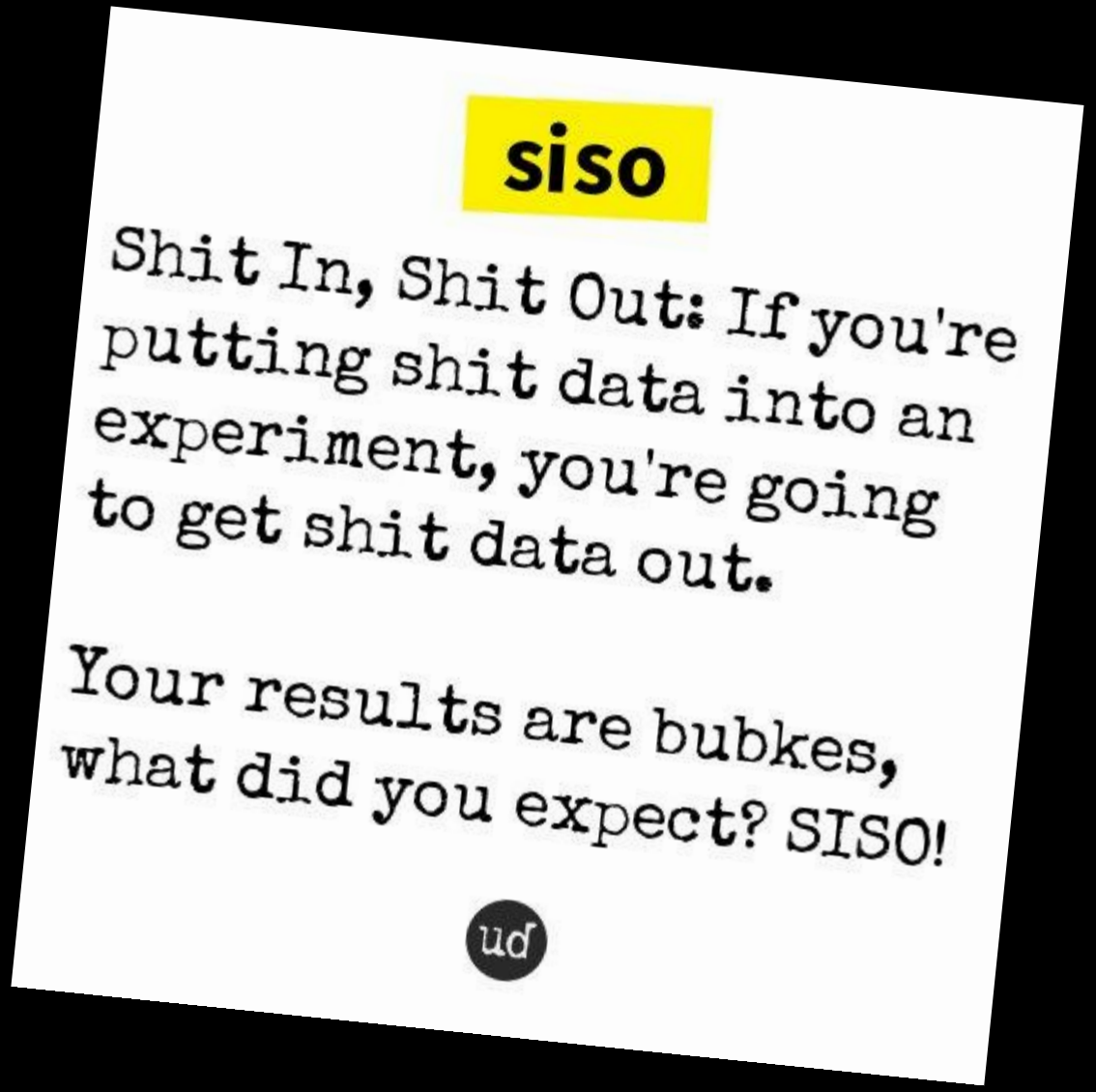


Image from Urban Dictionary: https://x.com/urbandictionary/status/958044898212630528

Diagnostics

“AI is wrong” to structured root cause analysis!

Where to start your investigation when the answer is not what you’re looking for

Permissions

- RLS / CLS
- Item level access
- governance policies

Prompt & grounding

- Clear question
- Proper metadata

Source grounding

- Semantic model, lakehouse, warehouse, eventhouse
- Relationships and item naming

Notes for model

- Tone & voice
- Context (what is a good performer, high or low?)
- Hints (example queries)

Generated query analysis

- Diagnostics

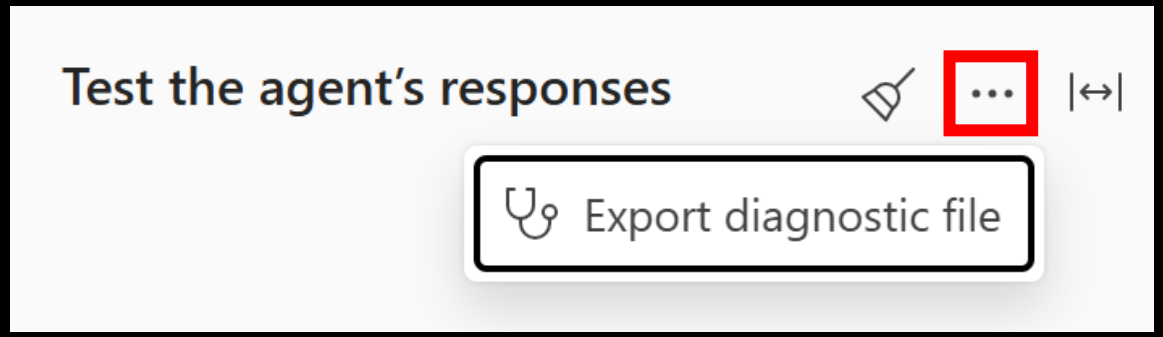
Capacity & performance

- Storage mode (for semantic models)
- Model size
- Load on the capacity

Diagnostics

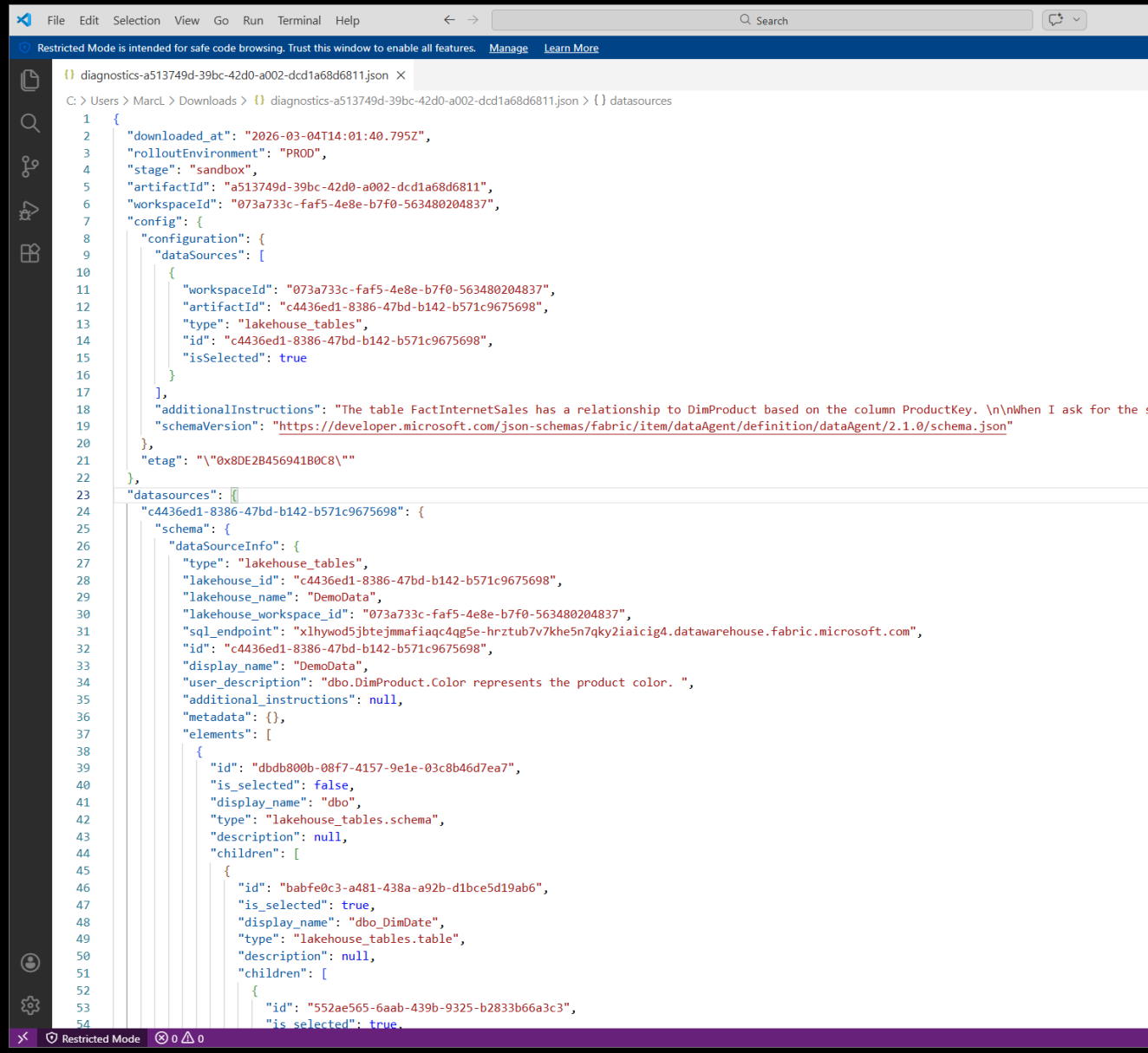
“AI is wrong” to structured root cause analysis!

Where to start your investigation when the answer is not what you’re looking for?



- User prompts
- Data source schemas and configurations
- File metadata and contents
- Environment and artifact identifiers (GUIDS)
- Any proprietary information you may have entered

Demo
Data Agent Diagnostics

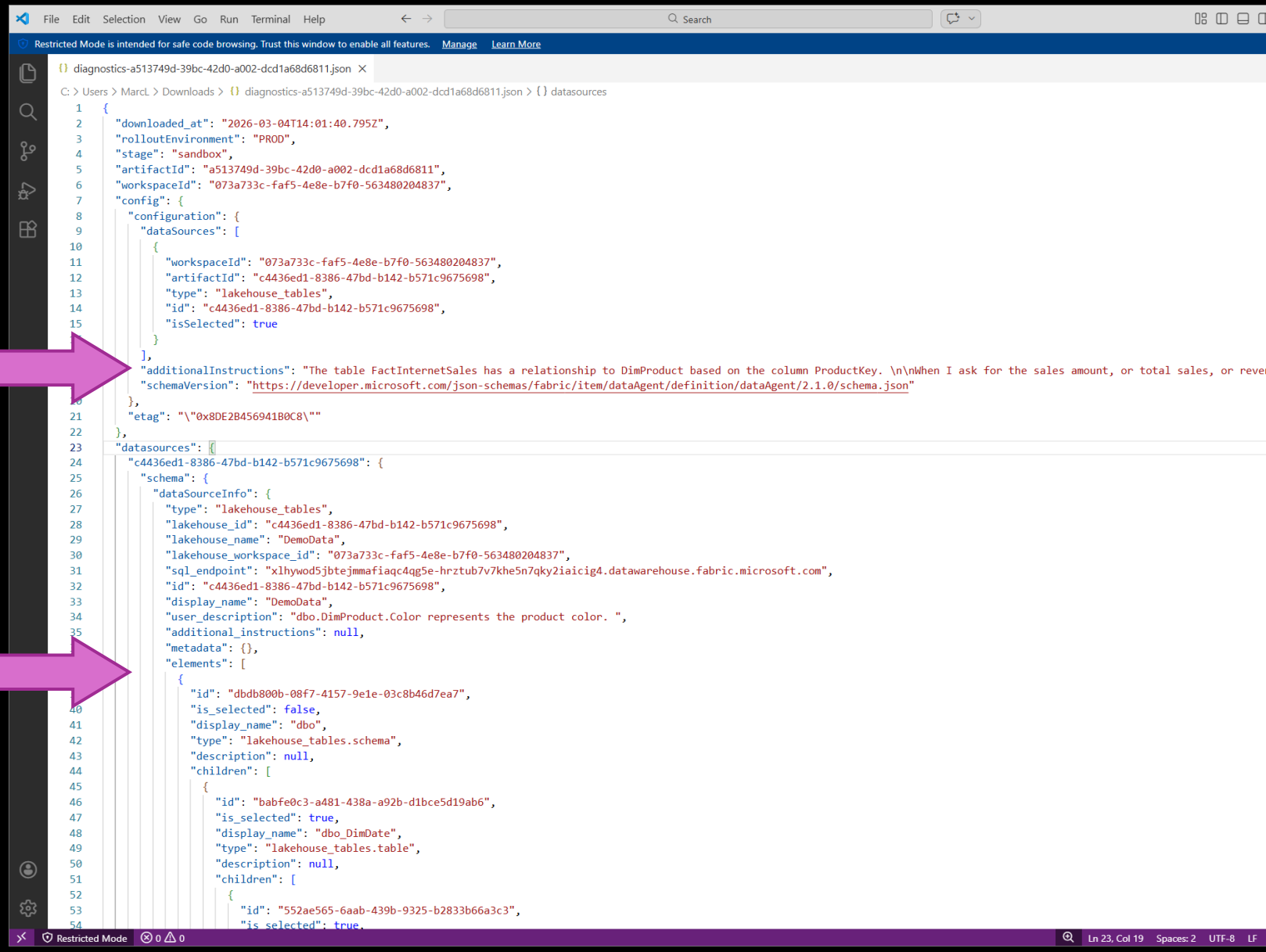


Diagnostics

Source instructions

Schema information

...and much more!





Data Agents
Software Development Kit

Data Agent SDK

Programmatic Management: Create, update, and delete Data Agent artifacts seamlessly.

Data Source Integration: Easily connect to and integrate multiple data sources, for enhanced data analysis and insight generation.

OpenAI Assistants API Support: Use the OpenAI Assistants API for rapid prototyping and experimentation.

Workflow Automation: Automate routine tasks to reduce manual efforts and improve operational efficiency.

Resource Optimization: Optimize configuration and management of Data Agent resources to better align with specific customer needs.

Copy

```
%pip install fabric-data-agent-sdk
```

Data Agent testing & validation

Gains confidence in the data agent, produces accurate, explainable and repeatable answers

- Establishes trust by validating your Data Agent against a known, authoritative dataset
- Results are compared one-to-one
- Enables objective evaluation

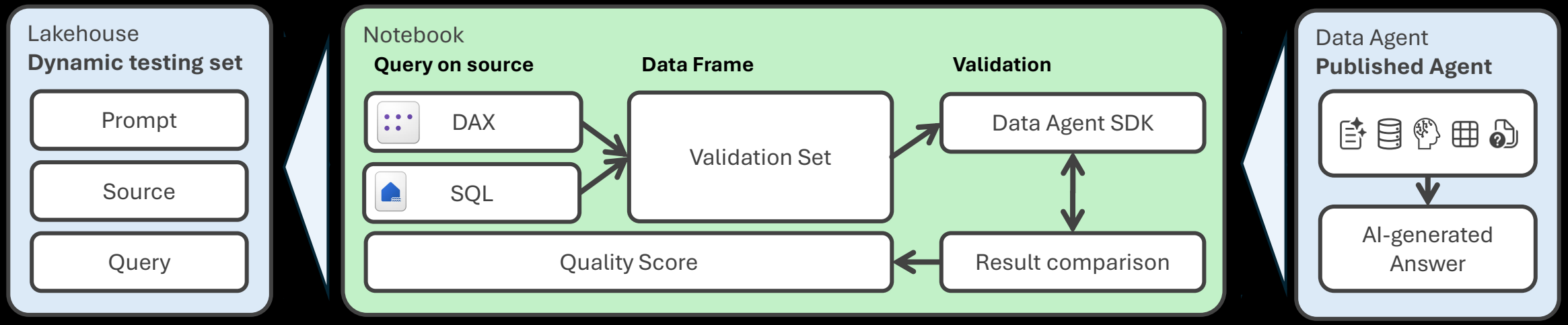
Static / fixed testing

- Predefined values stored in the notebook (dataframe) for testing
- Easy to setup and guarantees results across runs
- Limited flexibility: cannot cover new or changing data
- Mainly used for unit tests or small-scale validations

Dynamic testing

- Stores the query instead of hardcoded values
- Pulls data from the source at runtime for realistic testing
- Enables validation against changing data
- Supports end-to-end checks
- Scales easily with multiple tables or environments, aligning with automated QA pipelines

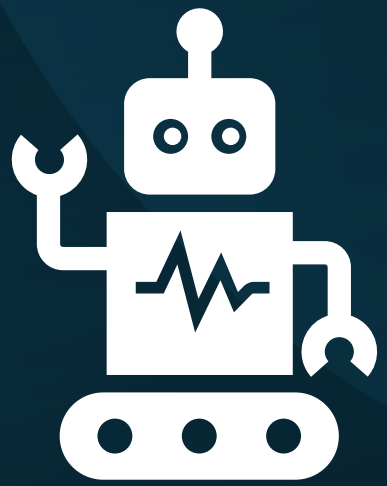
Dynamic testing



Specifics used:

Fabric Semantic Link for executing DAX on a semantic model

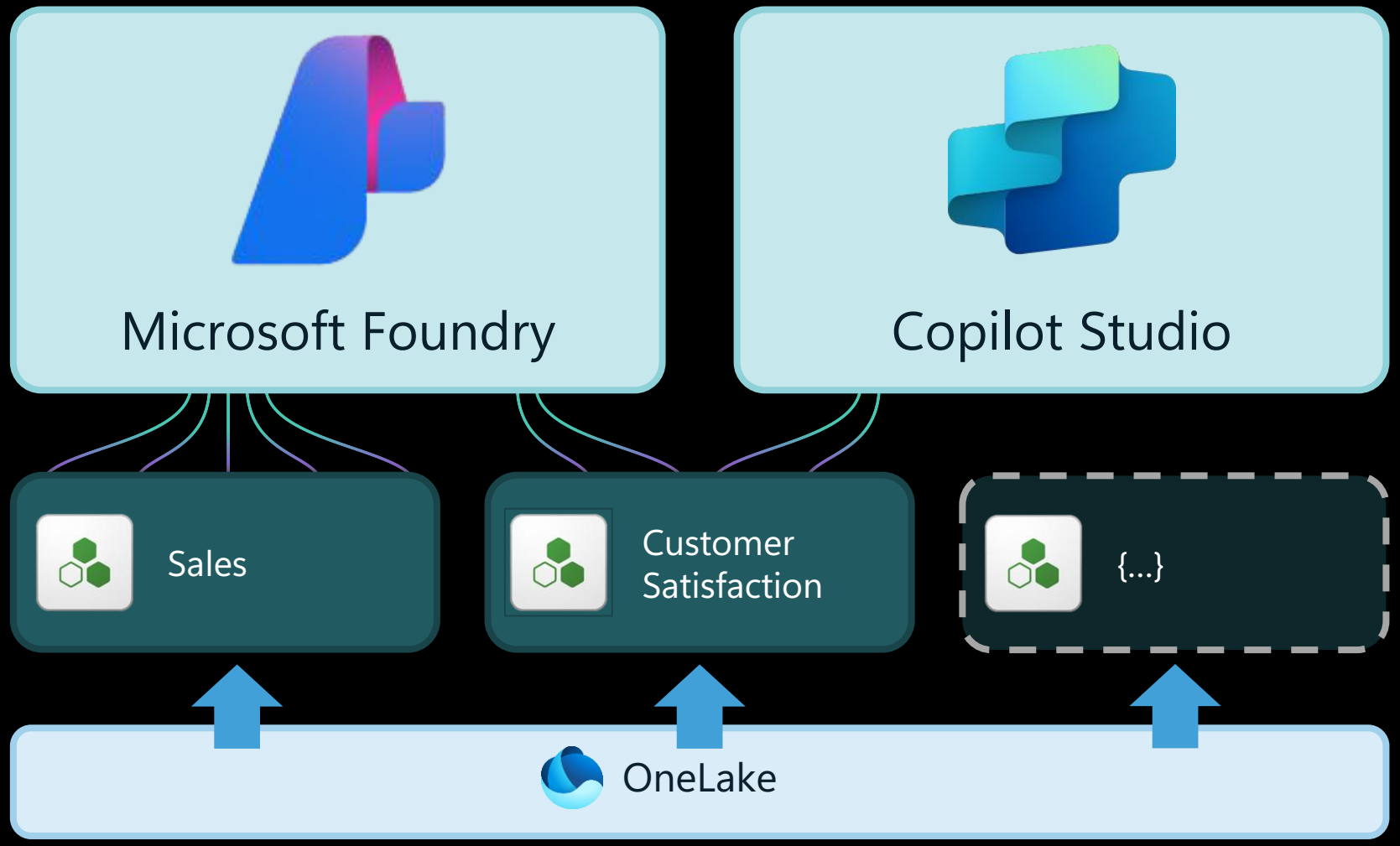
Data Agents SDK connecting and validating prompts in a Data Agent



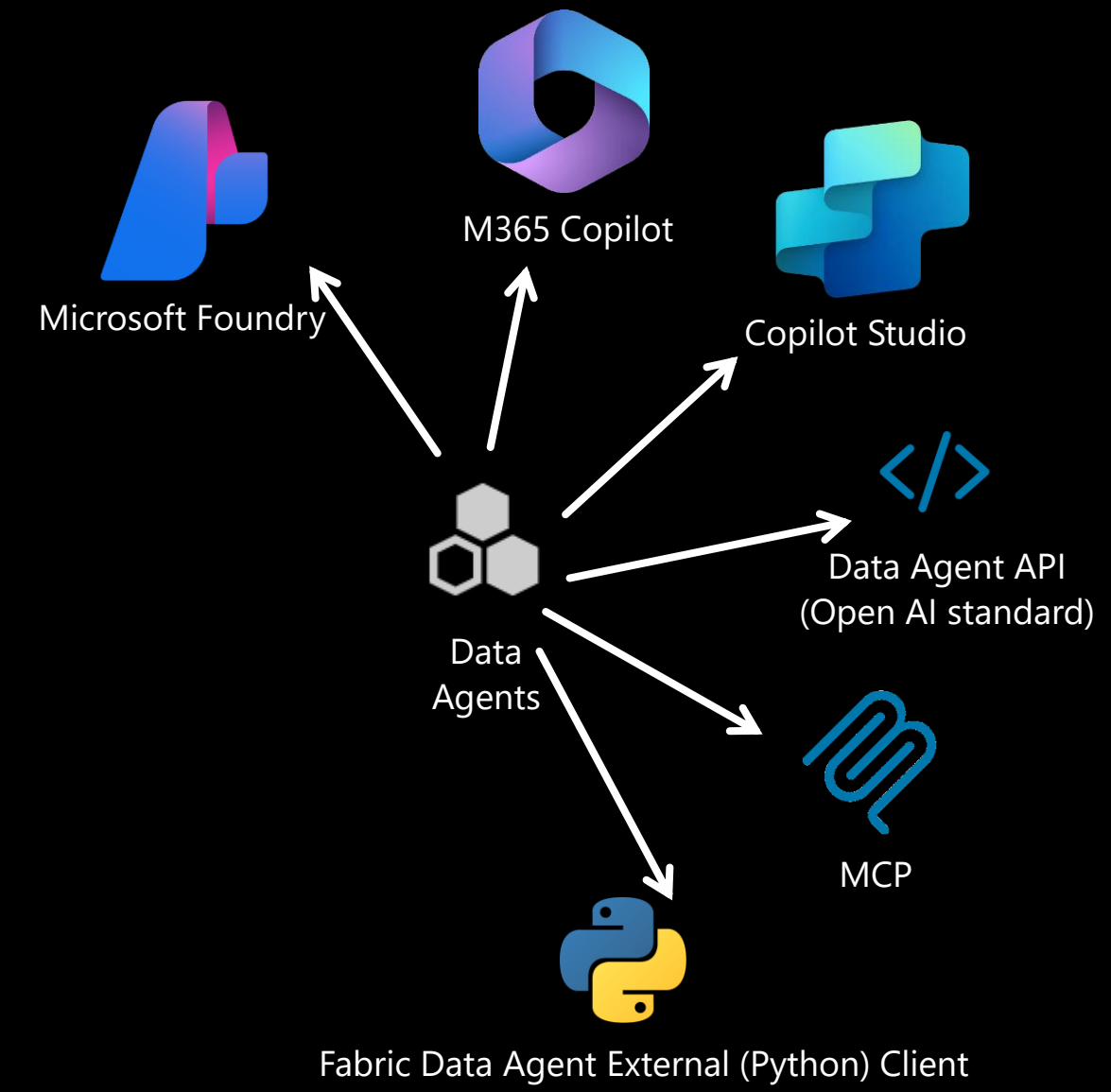
Multi-Agent **Architecture**

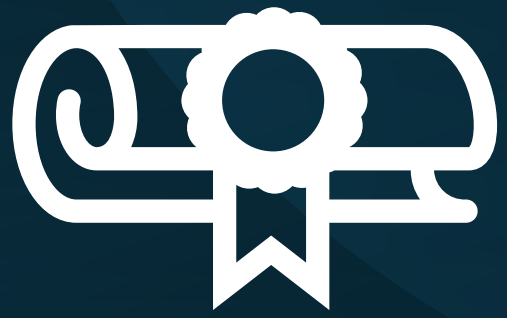
Agent Architecture

One to rule them all?



Part of a larger **ecosystem**





Wrap-up

Recipe to successful implementation



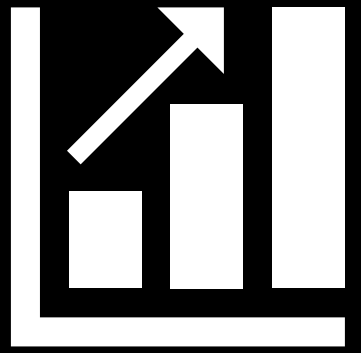
Reliable answers

Through decent preparation & (automated) testing



Secured

Inheritance and passthrough of permissions



Well adopted

Measured and tested with diagnostics

Wrap-up

- Fabric Data Agents unlock **governed access to enterprise data**, enabling users to ask questions in plain language across OneLake, semantic models, lakehouses, warehouses, and eventhouses.
- **Strong foundations matter most**, with well-prepared schemas, verified answers, and clear AI instructions driving accuracy and trust.
- **Security and governance are built-in**, with pass-through permissions, RLS/OLS inheritance, and per-user evaluation ensuring answers are both secure and compliant.
- Diagnostics, testing, and validation turn “AI is wrong” into insight, enabling structured root-cause analysis, quality scoring, and continuous **improvement through SDK-driven automation**.
- Data Agents fit into **a broader ecosystem**, integrating with Fabric, Power BI, M365, Copilot Studio, Microsoft Foundry, and custom applications.

Key takeaway:
Reliable AI answers come from reliable data, thoughtful design, and continuous validation.

Relevant Resources

Data Agent tenant settings
https://learn.microsoft.com/en-us/fabric/data-science/data-agent-tenant-settings

Data Agent sharing
https://learn.microsoft.com/en-us/fabric/data-science/data-agent-sharing

Data Agent SDK
https://learn.microsoft.com/en-us/fabric/data-science/fabric-data-agent-sdk

Data Agent Evaluation (testing)
https://learn.microsoft.com/en-us/fabric/data-science/evaluate-data-agent

Data Agent Best Practices
https://learn.microsoft.com/en-us/fabric/data-science/data-agent-configuration-best-practices

Semantic Model Best Practices for Data Agents
https://learn.microsoft.com/en-us/fabric/data-science/semantic-model-best-practices

Thanks for attending!



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Mastering Fabric Data Agents: From Setup to Success

Session Evals

These slides + the code
https://github.com/marclelijveld/Slide-decks
https://github.com/marclelijveld/fabric-automation